

GARV SHARMA

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PROFESSIONAL SUMMARY

Computer science engineer with strong data structures and algorithms fundamentals (Codeforces Pupil, CodeChef 3-Star) and hands-on experience across the stack —Java/Spring Boot and Node.js on the backend, React.js on the frontend, plus Python for applied ML. Comfortable picking up new tools quickly and shipping working code in fast-moving, collaborative teams.

WORK EXPERIENCE

IT Solutions Intern

May 2026 – July 2026

Sutherland Global Services

- Analyzed enterprise RFPs/RFIs across six service lines (Application Development, Cloud, ERP, Cybersecurity, AI & Data Analytics, Digital Transformation), translating business requirements into solutioning input for enterprise engagements.
- Evaluated enterprise IT landscapes and delivery models to assess alignment between customer objectives, technical feasibility, and commercial strategy.
- Studied three commercial engagement models (T&M, Fixed Price, BOT) and AI-driven solutioning approaches through structured knowledge-transfer sessions, working closely with pre-sales professionals to connect business goals to technical delivery

Cryptography Intern

December 2024 – January 2025

Defence Research and Development Organization (DRDO)

- Built a secure peer-to-peer chat application using AES symmetric encryption to protect message and file transfers, achieving end-to-end confidentiality.
- Partnered with 2 senior researchers on cryptographic key management, improving protocol robustness against replay and man-in-the-middle attacks by 40%, validated through 200+ simulated intrusion tests.
- Authored 10+ system architecture and threat model documents, expanding the team's secure-communications knowledge base by 40% and cutting security review time from 5 days to 3.

PROJECTS

Simple Kafka: Distributed Commit-Log Engine | *Java, Java NIO, Apache ZooKeeper*

- Architected a distributed, partitioned, append-only commit-log engine from scratch in Java, implementing segmented disk storage and 16-byte sparse indexing for $O(\log N)$ message lookup.
- Built a non-blocking TCP network layer using Java NIO (ServerSocketChannel, ByteBuffer) with a custom binary wire protocol, replacing JSON/HTTP serialization overhead.
- Implemented cluster coordination via Apache ZooKeeper across a 3-broker cluster, including ephemeral broker registration, automated controller election, and automatic partition leadership reassignment on broker failure — verified through live failover testing without interrupting active consumers.

AI Predictive Maintenance Copilot | *Python, Scikit-learn, Streamlit, Plotly, Gemini API*

- Developed an end-to-end predictive maintenance tool combining Isolation Forest for anomaly detection and Random Forest for failure prediction, reaching 99.7% accuracy, 97% precision, and 85% recall after feature engineering.
- Engineered features from rolling averages, rolling standard deviations, and sensor telemetry to improve prediction reliability and model explainability.
- Designed an interactive Streamlit dashboard with fleet analytics and a Gemini-powered natural-language assistant, giving maintenance engineers a faster path from raw sensor data to a real-world decision.

Airbnb-Style Booking Platform | *Java, Spring Boot, React.js, PostgreSQL*

- Architected a full-stack booking platform with Spring Boot, React.js, and PostgreSQL, covering 50+ REST API endpoints for authentication, listings, bookings, and payments.

- Reduced API latency by ~20% and database query response time by ~30% through query optimization and schema redesign.

EDUCATION

Birla Institute of Technology, Mesra <i>B.Tech, Computer Science & Engineering CGPA: 7.71</i> Apeejay School — 12th Grade, Science (PCM): 76% Apeejay School — 10th Grade: 86%	September 2023 – September 2027 <i>Ranchi, Jharkhand</i> Noida, Uttar Pradesh — 2023 Noida, Uttar Pradesh — 2021
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LANGUAGES & TECHNOLOGIES

Data & Business Analytics: SQL, Python, Data Analysis, RFP/RFI & Requirements Analysis, Enterprise Solutioning Distributed Systems: Java NIO, Apache ZooKeeper, Custom Binary Protocols Databases: SQL, PostgreSQL, MongoDB, NoSQL Design Languages & Frameworks: C++, Java, JavaScript, React.js, Spring Boot Cloud & Core CS: Google Cloud Platform, AWS, DSA, OOP, OS, DBMS, Git
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CODING PROFILES

Codeforces — Pupil (1230)	CodeChef — 3 Star (1641)
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ACHIEVEMENTS & CERTIFICATIONS

- Smart India Hackathon 2024 (Lead Developer) — Runner-up among 75 teams (420+ participants); delivered a data-driven IoT proof-of-concept under a 36-hour Agile sprint.
- Ranked top 20 nationally (SOF IEO) and top 30 (SOF IMO) among 50,000+ participants.
- Certifications: Google AI Essentials, Google Cloud Study Jam (Cloud Computing Foundations), GenAI & RAG Bootcamp, 30 Days of Google Cloud.

LEADERSHIP

- **Technical Member**, Google Developer Group (GDG), BIT Mesra
- **Content Head**, Society for Data Science (SDS), BIT Mesra